

AirSense

Large-Scale Urban Air Quality Data Warehouse & Intelligent AQI Analytics Platform

1. Project Vision

AirSense is a citizen-facing environmental intelligence platform built on a large historical air-quality Data Warehouse and Machine Learning pipeline.

The system will transform large-scale hourly air-quality observations into:

- understandable pollution information,
- historical and multidimensional analytics,
- pollution pattern discovery,
- AQI forecasting,
- pollution anomaly detection,
- station/city clustering,
- feature-selection and dimensionality-reduction insights,
- and personalized environmental activity planning.

The central idea is:

Do not build an AQI prediction model and then attach a dashboard.

Build:

Large-scale data → Data Warehouse → OLAP/Data Mining → ML → Citizen Intelligence UI

The current XKDR India Air Quality Database provides approximately 196.5M hourly readings from 558 monitoring stations and 15 pollutants, with monthly Parquet bulk files and metadata for stations and parameters. XKDR documents the data as sourced from the CPCB Continuous Ambient Air Quality Monitoring network plus five US Embassy/Consulate monitors.

2. Real-World Problem

Citizens usually see an AQI number but lack context.

A person may want to know:

- What is the air quality near me?
- Which pollutant is driving the current condition?
- Is today's pollution normal for this place?
- Is pollution improving or worsening?
- What does the next few hours look like?
- Which nearby monitoring areas historically have cleaner air?
- Is the present pollution level an unusual event?
- What does the available air-quality information mean for a planned outdoor activity?

CPCB describes the National AQI as a public-information tool intended to communicate air-quality conditions, and the Indian AQI framework uses six categories from Good to Severe.

AirSense will therefore act as a **citizen-oriented environmental decision-support system**, not merely an AQI display.

3. Primary Users

Citizen

Uses the system to understand air quality around a chosen location and selected time/activity.

Student / Analyst

Uses the system to explore historical patterns, OLAP operations, clusters, anomalies, features and model behaviour.

DWM Evaluator

Uses the DWM section to inspect:

- ETL,
- warehouse structure,
- dimensions/facts,
- star schema,
- snowflake schema,
- OLAP,
- aggregation,
- data marts,
- preprocessing,

- data quality.

ML Evaluator

Uses the ML section to inspect:

- feature engineering,
 - feature selection,
 - PCA,
 - clustering,
 - regression,
 - classification,
 - forecasting,
 - anomaly detection,
 - model comparison,
 - model evaluation,
 - explainability.
-

4. Source Data

Primary dataset

XKDR India Air Quality Database

Current published characteristics:

- approximately 196.5M hourly readings,
- 558 monitoring stations,
- 15 pollutants,
- January 2009–March 2026 coverage,
- monthly Parquet bulk files,
- station metadata,
- pollutant metadata,
- no bulk row cap.

The public repository also provides a Google Colab quickstart notebook demonstrating one-year extraction, city comparisons, daily cycles and station maps.

Recommended acquisition method

Use:

Bulk Parquet → Colab/DuckDB → ETL → Warehouse

Do not depend on repeated API calls for the main data pipeline.

XKDR explicitly documents monthly files under paths such as:

`v1/measurements/year=2024/month=03/data.parquet`

and provides `/v1/files` to enumerate bulk files. It also recommends using bulk files instead of long-running API queries for large extracts.

Supporting resources

Use station metadata for:

- latitude,
- longitude,
- city,
- state,
- station ID.

Use pollutant metadata for:

- pollutant name,
- units,
- parameter identifiers.

XKDR documents these as dedicated metadata endpoints/files.

AQI reference

For Indian AQI interpretation/calculation, use the CPCB National AQI documentation.

CPCB's framework uses eight AQI pollutants:

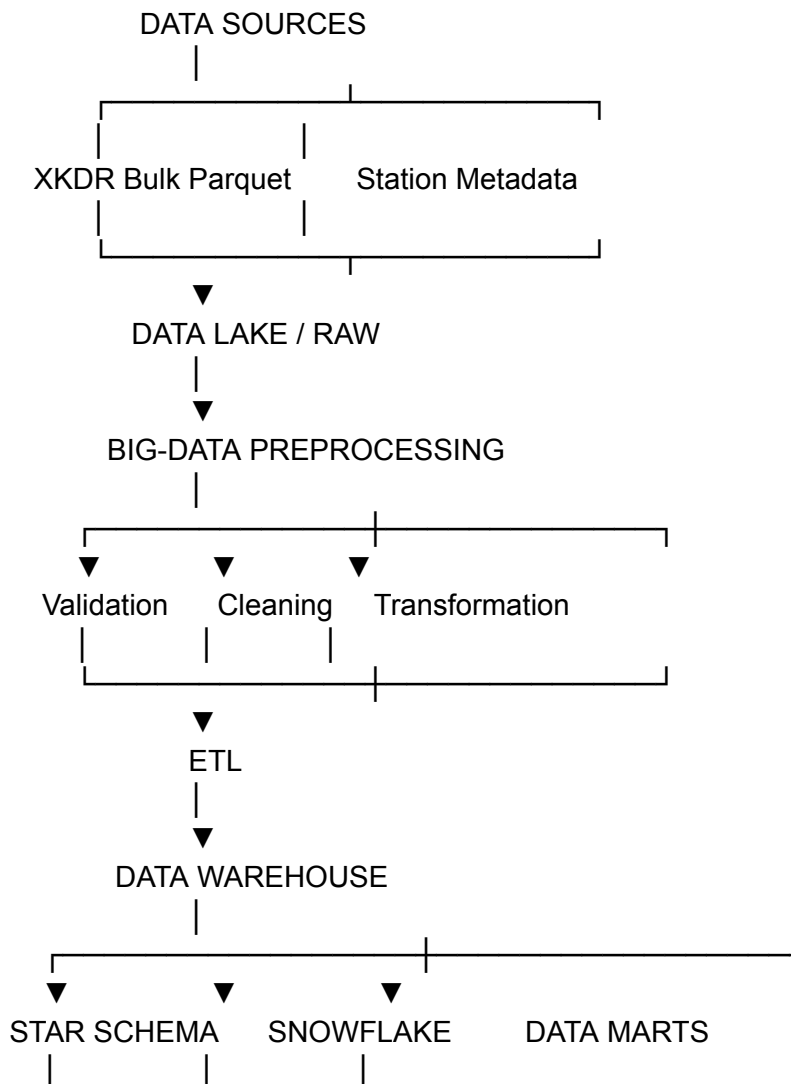
- PM10
- PM2.5
- NO2
- SO2
- CO
- O3
- NH3
- Pb

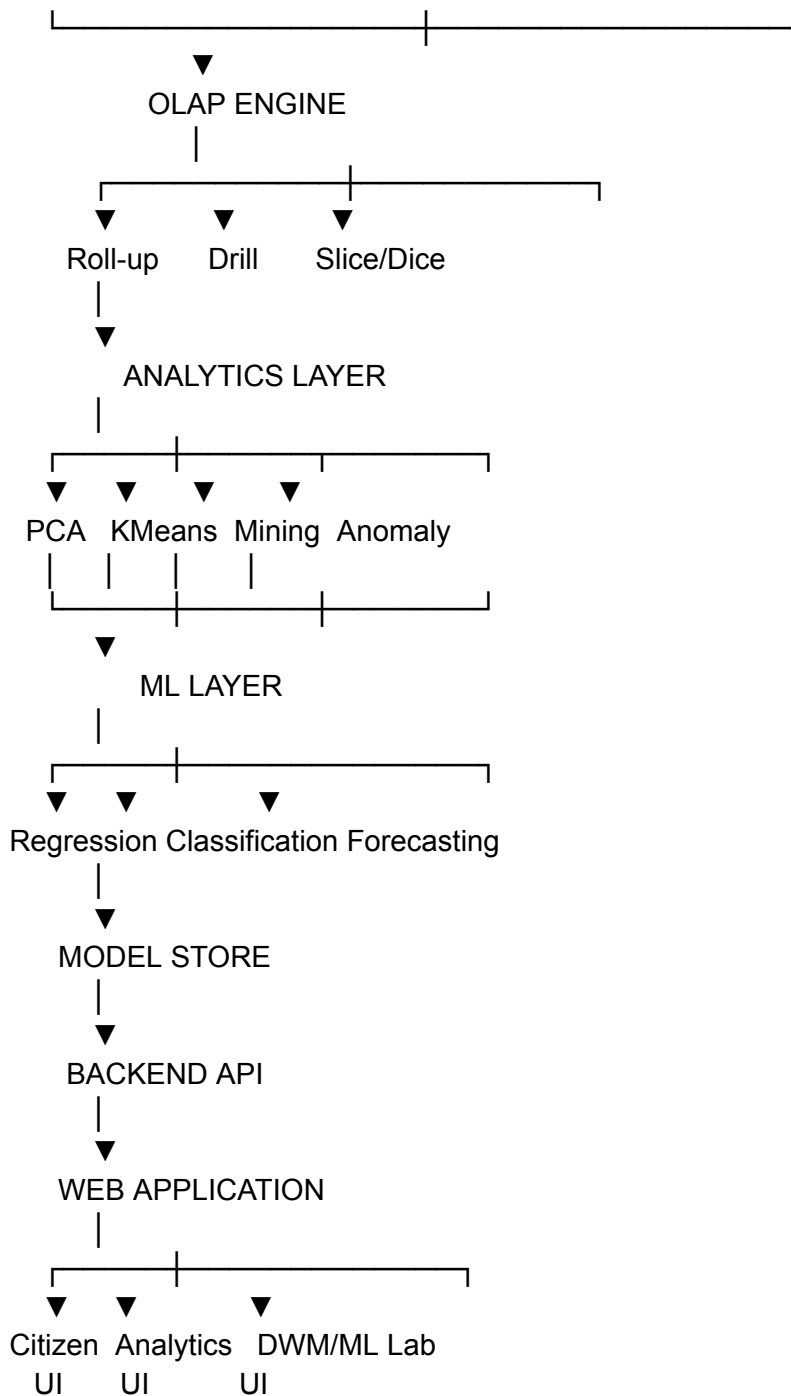
and six AQI categories:

- Good: 0–50
- Satisfactory: 51–100
- Moderate: 101–200
- Poor: 201–300
- Very Poor: 301–400
- Severe: 401–500.

The warehouse may retain all 15 measured pollutants while the AQI-calculation layer follows the CPCB-supported pollutant set and availability rules.

5. Complete Technical Architecture





6. Technology Stack

Data / DWM

Python

- Pandas
- NumPy
- PyArrow

DuckDB

Use DuckDB for large Parquet exploration, profiling and ETL. DuckDB can query Parquet directly and push filters/projections into the scan, making it suitable for large analytical files.

PostgreSQL

Use PostgreSQL as the main Data Warehouse.

Recommended advanced additions:

- partitioned fact tables,
- indexes,
- materialized views,
- aggregate tables.

PostgreSQL materialized views persist query results and can be refreshed, which is useful for dashboard/OLAP aggregates.

ML

- scikit-learn
- XGBoost
- optionally LightGBM
- optionally TensorFlow/PyTorch for an advanced forecasting model

Backend

FastAPI

Responsibilities:

- warehouse queries,
- OLAP APIs,
- model inference,
- analytics endpoints,
- location-to-nearest-station logic.

Frontend

Recommended:

- Next.js / React
 - TypeScript
 - Tailwind CSS
 - Leaflet or MapLibre for maps
 - Plotly/ECharts/Recharts for analytical graphics
-

7. Data Warehouse Model

The warehouse should use a **fact constellation** rather than forcing everything into one table.

Fact 1 — Fact_Air_Measurement

Grain

One pollutant measurement from one station at one timestamp.

Fields:

measurement_id

time_id

station_id

location_id

pollutant_id

source_id

measurement_value

quality_flag

This preserves the original analytical grain.

8. Fact 2 — Fact_Air_Daily

Grain

One station × one day.

Measures:

avg_pm25
max_pm25
avg_pm10
max_pm10
avg_no2
avg_so2
avg_co
avg_o3
avg_nh3

aqi
dominant_pollutant
anomaly_score

This becomes your primary BI/OLAP table.

9. Dimensions

Dim_Time

time_id
timestamp
date
hour
day
weekday
week
month
quarter
year
season
is_weekend
time_of_day

Hierarchy:

Year



Quarter



Month



Day



Hour

Dim_Station

station_id
station_name
latitude
longitude
city
state
station_type
source
status

Hierarchy:

India



State



City



Station

Dim_Pollutant

pollutant_id
pollutant_name
unit
pollutant_group

aqi_supported

Hierarchy:

Pollutants

↓

Particulate / Gaseous

↓

Specific Pollutant

Dim_Location

location_id

city

district

state

region

latitude

longitude

Dim_Source

source_id

source_name

network

country

10. Advanced DWM Features

These should be included where practical.

Star Schema

Primary operational/analytical model.

Snowflake Schema

Normalize:

Station



City



State

and:

Pollutant



Pollutant Group

SCD Type 2

Use for station metadata if station properties change.

Example:

Station A

Version 1 → Old city/metadata

Version 2 → Updated metadata

This gives you a concrete **Slowly Changing Dimension** example.

Fact Constellation

Two shared facts:

Fact_Air_Measurement

Fact_Air_Daily

using conformed dimensions.

Data Marts

Create:

Citizen Analytics Mart

ML Feature Mart

Air Quality BI Mart

11. ETL Pipeline

PARQUET



Extract



Schema Validation



Timestamp Validation



Station Validation



Duplicate Detection



Missing-Value Analysis



Outlier Detection



Unit Normalization



Feature Engineering



AQI Derivation



Dimension Generation



Surrogate Keys



Fact Generation



Warehouse Load

12. Preprocessing Requirements

The Colab notebook must explicitly demonstrate:

Missing-value analysis

Pollutant

Missing %

Longest gap

Station coverage

Duplicate analysis

Station + Timestamp + Pollutant

Outliers

Demonstrate:

- IQR
- Z-score
- domain-bound checks
- optionally Isolation Forest

Do not blindly remove all extremes.

The source is explicitly published as received from the monitoring networks, without XKDR performing gap filling, outlier removal or calibration. That makes your preprocessing/data-quality stage meaningful.

Temporal cleaning

Handle:

- duplicate timestamps,
- missing hours,
- inconsistent temporal sequences,
- station-specific gaps.

XKDR states that timestamps are expressed in IST, with CPCB readings typically on the hour and embassy readings at half past.

13. Feature Engineering

Create:

Temporal features

hour
day_of_week
month
quarter
season
is_weekend
rush_hour
time_of_day

Lag features

PM25_lag_1
PM25_lag_3
PM25_lag_6
PM25_lag_12
PM25_lag_24

Rolling features

PM25_rolling_3h
PM25_rolling_6h
PM25_rolling_12h
PM25_rolling_24h

Statistical features

rolling_mean
rolling_std
rolling_min
rolling_max
rate_of_change

Cross-pollutant features

PM25_PM10_ratio
NO2_CO_ratio
particulate_index
pollutant_count_above_threshold

Station-baseline features

current_value / historical_mean
deviation_from_station_mean
zscore_vs_station_history

These make your ML work significantly more interesting.

14. Feature Selection & Extraction Module

Create a dedicated notebook and UI page.

Feature Selection

Demonstrate:

Correlation filtering
Mutual Information
SelectKBest
Recursive Feature Elimination
Tree-based importance

UI:

FEATURE SELECTION

All Features: 32

Selected Features: 14

PM2.5	████████████████
PM10	██████████████
NO2	██████████
Hour	██████
Rolling_PM25	████████████████
...	

15. Feature Extraction

Use:

PCA

Original feature space



Standardization



PCA



PC1

PC2

PC3

Show:

- explained variance,
- cumulative explained variance,
- loading plot,
- 2D station projection,
- 3D projection.

This becomes both:

ML feature extraction

and

DWM/data-mining exploratory analysis.

16. Data Mining — Clustering

Do not make clustering an isolated lab result.

Use it to create **pollution profiles**.

Station clustering

Features:

avg_PM25
avg_PM10
avg_NO2
avg_SO2
avg_CO
avg_O3
pollution_variability
peak_frequency

Algorithms:

- K-Means
- Hierarchical Clustering
- DBSCAN

Evaluation:

- Elbow method
- Silhouette score
- Davies-Bouldin score

UI:

POLLUTION PROFILES

Cluster 1
Lower particulate profile

Cluster 2
High particulate profile

Cluster 3
Multi-pollutant profile

Cluster 4
High temporal variability

Descriptions should be generated from actual cluster statistics, not hard-coded assumptions.

17. Anomaly Detection

Purpose:

Find unusual pollution events.

Methods:

Z-score

IQR

Isolation Forest

DBSCAN

UI:

ANOMALY DETECTED

Station:

Mumbai-Andheri

Pollutant:

PM2.5

Observed:

...

Historical baseline:

...

Deviation:

...

Severity:

UNUSUAL

The system should show the evidence behind the anomaly rather than simply displaying "Anomaly = 1".

18. Correlation & Pollutant Relationship Explorer

Show:

Correlation Matrix

between:

PM2.5

PM10

NO2

SO2

CO

O3

NH3

...

Allow user to:

Select city

Select year

Select season

Then dynamically regenerate the correlation matrix.

This is a bridge between:

EDA + data mining + ML feature selection.

19. ML Task 1 — AQI Regression

Goal

Predict future AQI.

Recommended targets:

AQI +1 hour
AQI +3 hours
AQI +6 hours
AQI +12 hours
AQI +24 hours

Models:

Baseline
Linear Regression

Model 1
Random Forest

Model 2
Gradient Boosting

Model 3
XGBoost

Optional advanced model:

LSTM / GRU

Do not make deep learning mandatory.

20. ML Task 2 — AQI Classification

Predict:

Good
Satisfactory
Moderate
Poor
Very Poor
Severe

Models:

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost
- SVM

Metrics:

- Accuracy
- Precision
- Recall
- F1
- Macro F1
- Confusion Matrix
- ROC-AUC where appropriate

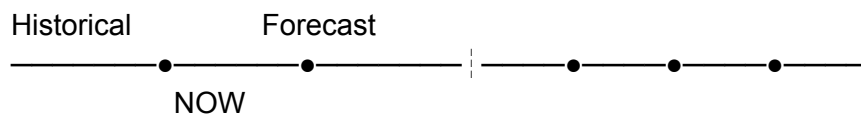
CPCB's AQI framework defines these six categories.

21. ML Task 3 — Pollution Forecast

The system should produce:

Next 1 hour
Next 3 hours
Next 6 hours
Next 12 hours
Next 24 hours

Graph:



Clearly distinguish observed data from predictions.

22. Model Comparison Page

This page is mandatory.

Section A — Metrics

MODEL COMPARISON

	MAE	RMSE	R ²	Inference
Linear	...	...	...	...
RF	...	...	...	...
XGBoost	...	...	...	...
GRU	...	...	...	...

Section B — Error plots

Actual vs Predicted

Residual distribution

Prediction timeline

Section C — Model selection rationale

The UI should automatically state something like:

SELECTED MODEL

XGBoost

Reason:

Lowest validation RMSE

+

Competitive MAE

+

Stable validation performance

+

Acceptable inference cost

Do not select a model merely because it has the highest one-off metric.

For forecasting, use **chronological train/validation/test splits** or **TimeSeriesSplit**, not a random shuffle that lets future information leak into training.

23. Model Explainability

Include:

Global feature importance

and optionally:

SHAP summary

SHAP dependence

Single-prediction explanation

Citizen-facing version:

WHY IS THE FORECAST HIGH?

PM2.5 recent trend	
PM10	
NO2	
Hour	
Previous AQI	

Analyst-facing version:

SHAP contribution

Feature

Value

Direction

Magnitude

24. Main Citizen Dashboard

This is the **actual product**.

User input

LOCATION

Use My Location

or

Search Manually

or

Select on Map

TIME

Use Current Time

or

Choose Date & Time

ACTIVITY

Walking

Running

Cycling

Outdoor Sports

Commute

Other

Location permission

When user clicks:

Use My Location

request browser geolocation permission.

Browser geolocation requires explicit user permission and normally HTTPS.

Use latitude/longitude to determine:

Nearest Station

City

State

Distance to Station

Time

Use:

Device/browser current time

Device/browser timezone

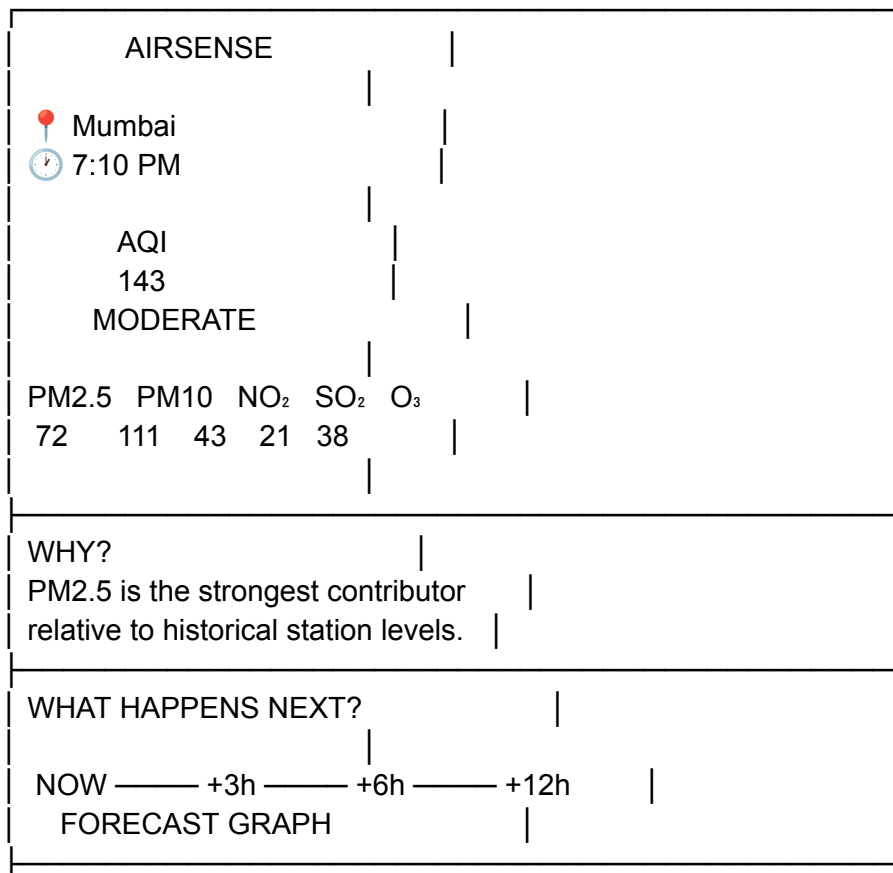
and allow:

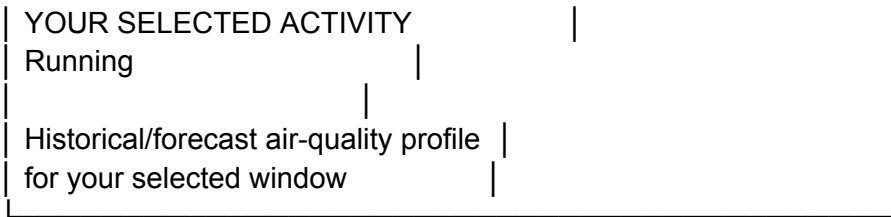
Manual date/time selection

Do not represent this as "GPS time."

The browser environment supplies local time/timezone information.

25. Citizen Dashboard Output





The dashboard must always distinguish **measured current/historical observations** from **ML predictions**.

26. Pollution Map

Dedicated interactive map.

Station markers

Every station is represented as:

CircleMarker

with radius based on:

zoom

station importance

and displayed state based on pollution category.

Conceptually:

Green → lower category

Yellow → higher

Orange → higher

Red → poor

Dark red → very poor/severe

Do not make the colors imply a medical diagnosis.

Clicking a station opens:

Station Name
Current/latest available reading
AQI
PM2.5
PM10
NO2
Historical average
Cluster
Anomaly status
Forecast

27. Map Modes

Provide a layer selector:

AQI
PM2.5
PM10
NO2
SO2
O3
Anomalies
Clusters

Example:

☒ AQI
☐ PM2.5
☐ PM10
☐ Clusters
☐ Anomalies

28. Pollution Heatmap

Map mode:

AREA POLLUTION DENSITY

Use station measurements to create a spatial visualization.

Important:

Do not pretend this is a directly measured continuous surface everywhere.

Label it clearly as:

"Station-based pollution visualization"

or:

"Interpolated/estimated pollution surface"

depending on the method actually implemented.

29. Historical Analytics Page

This is the citizen/analyst bridge.

Features:

City comparison

Mumbai
Delhi
Pune
Bengaluru
Hyderabad

Historical trend

2009 ————— 2026

Seasonal comparison

Winter
Summer
Monsoon
Post-monsoon

Weekday vs weekend

Weekday

Weekend

Hourly pollution clock

00 04 08 12 16 20 24



30. "Why Is My Air Bad?" Page/Panel

User chooses:

Location

Date

Time

System shows:

Current level

Historical baseline

Deviation

Dominant pollutant

Top contributing features

Example structure:

CURRENT

PM2.5 = ...

NORMAL FOR THIS TIME

PM2.5 = ...

DEVIATION

+32%

MAIN DRIVER

PM2.5

OTHER CONTRIBUTORS

NO2

PM10

31. "Is This Normal?" Feature

This is one of the strongest citizen features.

Compare current conditions against:

Same station

Same month

Same day-of-week

Same hour

Historical baseline

Output:

TODAY VS TYPICAL

PM2.5

Current: ...

Typical: ...

Difference: ...

Status:

WITHIN NORMAL RANGE

or

UNUSUAL

This relies heavily on the warehouse's historical dimensions.

32. "Find a Better Time" Feature

User chooses:

Activity:
Running

Available window:
5 PM – 10 PM

System shows:

5 PM
AQI ...

6 PM
AQI ...

7 PM
AQI ...

8 PM
AQI ...

9 PM
AQI ...

10 PM
AQI ...

Show the underlying:

- observed history where available,
- forecast values for future periods,
- confidence/error information.

Avoid claiming personalized medical safety.

33. "Find a Better Area" Feature

User chooses:

City
Activity
Time range

The system compares nearby monitored areas using:

Historical mean
Recent conditions
Forecast
Pollutant profile
Anomaly rate

Map:

Area A •
Area B •
Area C •
Area D •

Clicking a location provides the analytical evidence.

34. DWM Showcase Page

This is specifically for **Ms. Shruti**.

Page title:

Data Warehouse Architecture

Sections:

Source layer

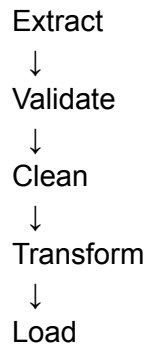
XKDR



Parquet

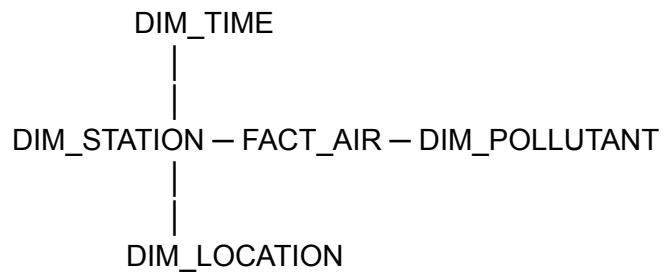
ETL

Animated flow:



Star Schema

Interactive diagram:

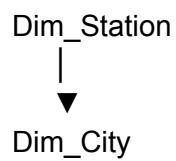


Click a table:

columns
data types
PK
FK
record count

35. Snowflake Schema Page

Show:



↓
▼
Dim_State

and:

Dim_Pollutant
↓
▼
Dim_Pollutant_Group

Allow toggling:

STAR
SNOWFLAKE

so the evaluator immediately sees the distinction.

36. OLAP Explorer

This deserves a dedicated interactive page.

Controls

Measure
[Average PM2.5 ▼]

Dimension
[Time ▼]

Hierarchy
[Year → Month → Day → Hour ▼]

Operation
○ Roll-up
○ Drill-down
○ Slice
○ Dice
○ Pivot

Example

Roll-up:

Station



City



State

Drill-down:

Year



Month



Day



Hour

Slice:

Pollutant = PM2.5

Dice:

Mumbai + Pune

2024–2026

PM2.5 + PM10

Winter

Pivot:

Cities × Months

This page should be backed by real warehouse SQL, not mocked frontend data.

37. DWM Concepts Used Page

Display a checklist with status:

- ✓ Data Cleaning
- ✓ Data Integration
- ✓ ETL
- ✓ Data Warehouse
- ✓ Fact Tables
- ✓ Dimension Tables
- ✓ Star Schema
- ✓ Snowflake Schema
- ✓ Hierarchies
- ✓ Surrogate Keys
- ✓ SCD
- ✓ Fact Constellation
- ✓ Data Marts
- ✓ Aggregation
- ✓ OLAP
- ✓ Roll-up
- ✓ Drill-down
- ✓ Slice
- ✓ Dice
- ✓ Pivot
- ✓ Visualization
- ✓ Clustering
- ✓ Anomaly Detection

Each concept should have:

Definition

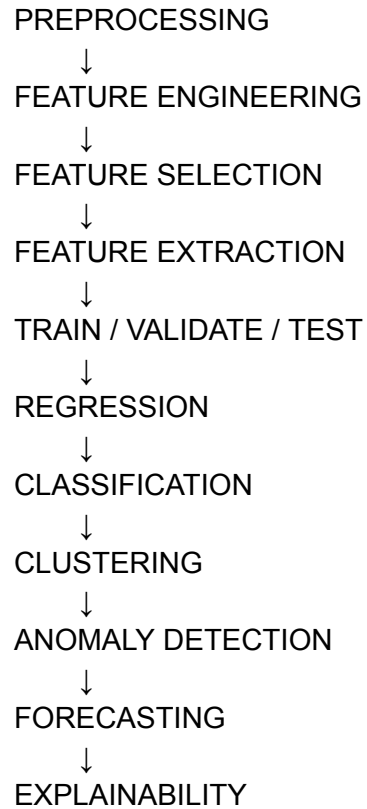
Where AirSense uses it

Live example

Query/result

38. ML Concepts Page

Show:



Every concept should open a small interactive demonstration.

39. Feature Lab Page

Tabs:

Raw Features
Engineered Features
Selected Features
PCA Components

Show:

32 original features

↓
Feature selection
↓

18 retained



PCA



8 principal components

Visuals:

- correlation heatmap,
 - feature importance,
 - mutual information chart,
 - PCA explained variance curve,
 - PCA scatter plot.
-

40. Data Visualization Page

Every chart should come from the warehouse or an explicitly documented analytical view.

Include:

KPI cards

Stations

Pollutants

Observations

Years

Cities

Anomalies

Visualizations

Time-series

Heatmap

Histogram

Box plot

Scatter plot

Correlation matrix

Station map

Cluster plot

Seasonal chart

Weekday/weekend chart
Pollution clock
City comparison

41. Data Quality Page

This is a strong DWM presentation feature.

Show:

RAW DATA

196.5M observations
558 stations
15 pollutants

QUALITY

Missing %
Duplicate %
Invalid %
Outlier %
Final usable records

Also show:

Station coverage
Pollutant coverage
Year coverage

42. ETL Monitoring Page

Visual pipeline:

SOURCE



INGESTION



VALIDATION



CLEANING



TRANSFORMATION



WAREHOUSE LOAD



AGGREGATION



ML FEATURE MART



Display:

Records processed

Records rejected

Processing duration

Rows inserted

Rows updated

43. Model Repository Page

Display all trained models.

AQI Regression

XGBoost

Version 1.2

MAE:

RMSE:

R²:

Status:

Production

AQI Classification

Random Forest

F1:

Precision:

Recall:

This demonstrates reproducibility.

44. Colab-to-UI Integration Requirement

This is **critical**.

The Colab notebook must not become a separate thing that the UI ignores.

Every major Colab result should produce a persistent artifact.

Colab output examples

cleaned_data.parquet
feature_dataset.parquet
station_clusters.csv
pca_components.csv
model_metrics.json
feature_importance.json
forecast_results.parquet
anomalies.parquet
warehouse_load_report.json

Then:

COLAB

↓

ARTIFACTS

↓

WAREHOUSE / MODEL STORE



FASTAPI



NEXT.JS

This directly satisfies your requirement that **the majority of the work performed in Colab must be visible in the final application.**

45. Recommended Colab Notebook Structure

Notebook 01 — Data Acquisition

Dataset discovery

Metadata

Parquet files

Coverage

Schema

Sampling

Notebook 02 — Large-Scale Data Exploration

DuckDB

Partition inspection

Row counts

Station counts

Pollutant counts

Temporal coverage

DuckDB can read multiple Parquet files directly and process them with predicate/projection pushdown.

Notebook 03 — Data Preprocessing

Missing values

Duplicates

Outliers
Validation
Cleaning

Notebook 04 — EDA

Time trends
Station analysis
Pollutant analysis
Correlation
Seasonality

Notebook 05 — Feature Engineering

Lag
Rolling
Temporal
Baseline
Interaction features

Notebook 06 — Feature Selection & PCA

Correlation
Mutual Information
RFE
PCA
Variance

Notebook 07 — Clustering

K-Means
Elbow
Silhouette
DBSCAN
Cluster profiles

Notebook 08 — Anomaly Detection

IQR

Z-score
Isolation Forest
Anomaly analysis

Notebook 09 — AQI Regression

Feature set
Train
Validation
Models
Metrics

Notebook 10 — AQI Classification

Categories
Models
Confusion matrices
Metrics

Notebook 11 — Forecasting

Lag model
Tree model
Advanced model
Future predictions

Notebook 12 — Explainability

Feature importance
SHAP
Error analysis

Notebook 13 — Warehouse / ETL

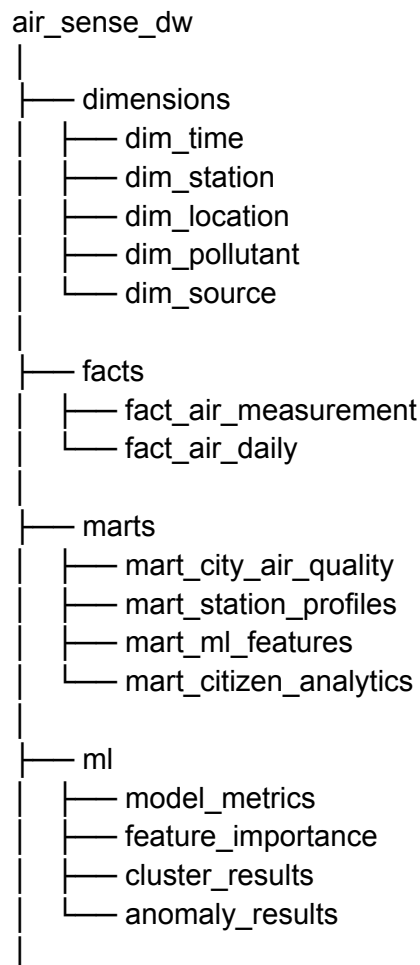
Dimensions
Facts
Surrogate keys
Star schema
Snowflake

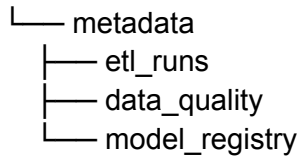
Load
Validation

Notebook 14 — Final Export

ML artifacts
Metrics
Charts
Feature tables
Cluster tables
Anomaly tables
API-ready data

46. Recommended Database Structure





47. Backend API Requirements

FastAPI endpoints should roughly include:

GET /api/location
GET /api/stations
GET /api/air/current
GET /api/air/history
GET /api/air/forecast
GET /api/air/anomalies
GET /api/air/clusters
GET /api/air/compare
GET /api/air/explain
GET /api/olap
GET /api/schema/star
GET /api/schema/snowflake
GET /api/features
GET /api/models
GET /api/data-quality
GET /api/etl

ML endpoints:

POST /api/predict/aqi
POST /api/predict/category
POST /api/anomaly/check
POST /api/activity-analysis

48. Complete UI Page Structure

/

Home / Landing

Contains:

- project introduction,
- real-world problem,
- simple explanation of AQI,
- what PM2.5 means,
- what PM10 means,
- what NO₂ means,
- what µg/m³ means,
- what "Good/Moderate/Poor" means,
- how AirSense works,
- data-source attribution,
- simplified architecture,
- "Explore Air Quality" CTA.

The language must be **normal-user friendly**.

Example:

PM2.5

Tiny particles suspended in air.

µg/m³

Micrograms of particles per cubic metre
of air.

Higher concentration generally means
more particulate pollution.

No unnecessary technical jargon on the landing page.

/dashboard

Personal Air Intelligence

Contains:

- location selector,
 - permission-based "Use My Location",
 - manual location,
 - current/selected date-time,
 - activity selector,
 - current air information,
 - forecast,
 - dominant pollutant,
 - "Why?",
 - "Is this normal?",
 - activity-window analysis.
-

/map

Pollution Map

Contains:

- station markers,
 - pollution categories,
 - map filters,
 - pollutant selector,
 - anomaly layer,
 - cluster layer,
 - station popup,
 - nearest-station identification.
-

/analytics

Historical Air Analytics

Contains:

- trends,
- city comparison,
- station comparison,
- seasonal analysis,

- hourly cycles,
 - weekday/weekend,
 - pollution calendar,
 - heatmaps.
-

/warehouse

Data Warehouse

Contains:

- architecture,
 - star schema,
 - snowflake schema,
 - fact tables,
 - dimension tables,
 - grain,
 - keys,
 - row counts,
 - relationships.
-

/olap

OLAP Explorer

Contains:

- roll-up,
 - drill-down,
 - slice,
 - dice,
 - pivot,
 - filters,
 - hierarchy selectors,
 - generated tables/charts,
 - optional SQL preview.
-

/datamining

Data Mining Lab

Contains:

- correlation,
 - clustering,
 - PCA,
 - anomaly detection,
 - pollution profiles,
 - cluster metrics,
 - interactive plots.
-

/features

Feature Engineering & Selection

Contains:

- raw features,
 - engineered features,
 - correlation filtering,
 - mutual information,
 - RFE,
 - feature importance,
 - PCA,
 - explained variance.
-

/models

ML Model Lab

Contains:

- regression models,
- classification models,
- forecasting models,

- metrics,
 - actual-vs-predicted,
 - confusion matrices,
 - residuals,
 - model comparison.
-

/explain

Explain My AQI

Contains:

- predicted AQI,
 - contributing variables,
 - baseline deviation,
 - feature importance,
 - historical comparison.
-

/pipeline

ETL & Data Quality

Contains:

- ETL flow,
 - processing counts,
 - missing-value statistics,
 - duplicates,
 - outliers,
 - rejected records,
 - warehouse load status.
-

/concepts

DWM + ML Concepts

Contains:

DWM
Preprocessing
ETL
Warehouse
Star
Snowflake
OLAP
Data Mining
Clustering
Visualization

ML
Regression
Classification
PCA
Feature Selection
Feature Extraction
Forecasting
Anomaly Detection
Explainability

Each concept must link to its actual implementation in the project.

/about

Contains:

- team,
 - project objective,
 - architecture,
 - sources,
 - methodology,
 - technology stack,
 - limitations,
 - citation/attribution.
-

49. Citizen-Friendly Information Design

The landing page must explain terms such as:

AQI
PM2.5
PM10
NO₂
SO₂
O₃
CO
NH₃
µg/m³
Monitoring station
Historical baseline
Forecast
Anomaly

Example:

AQI
Air Quality Index.
A simplified indicator used to communicate
air-quality conditions.

PM2.5
Very small airborne particles with an
aerodynamic diameter of 2.5 micrometres or less.

µg/m³
Micrograms per cubic metre of air.

Use CPCB terminology for AQI categories.

50. Privacy Requirements

Location data should be:

- requested only when needed,

- explicitly permission-based,
- optionally entered manually,
- processed only for the user's request where possible,
- not permanently stored unless the user explicitly opts in.

Fallback:

Location permission denied



Search city



Select station



Continue

This ensures the app remains usable on desktop/laptop.

51. Important Data/Health Disclaimer

The UI must distinguish:

Environmental information

from:

medical advice.

KXDR explicitly warns that its compiled data is provided as-is and has not been validated by XKDR for regulatory, legal or health decisions.

Therefore the application should use wording such as:

"AirSense provides environmental information and model-based estimates for informational purposes."

rather than:

"AirSense determines whether it is medically safe for you to exercise."

52. Advanced Big-Data Demonstration

Because the source is genuinely large, demonstrate:

196.5M source observations
↓
Partitioned Parquet
↓
DuckDB scan
↓
Selective columns
↓
Filtered time/location slices
↓
Aggregated analytical datasets

DuckDB supports direct Parquet querying, parallel processing and filter/projection pushdown.

Do not load the entire 196M-row dataset into Pandas at once.

Use:

DuckDB
+
PyArrow
+
Parquet

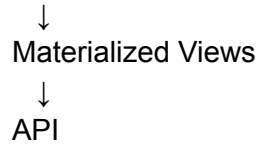
for large-scale processing.

53. Performance Requirements

Dashboard should not query the raw 196M-row fact every time.

Use:

Raw Fact
↓
Daily Aggregates



PostgreSQL materialized views are appropriate for persisting expensive aggregate query results and refreshing them when required.

Target:

Dashboard KPI: <2 seconds

OLAP query: <3 seconds

Map rendering: <3 seconds

Model inference: <2 seconds

These are project targets, not claims about guaranteed performance.

54. What Each Team Member Can Own

Member 1 — DWM/Data Engineering

Data acquisition
Parquet
DuckDB
Preprocessing
ETL
PostgreSQL
Star schema
Snowflake
OLAP
Data quality
Materialized views

Member 2 — ML/Data Mining

Feature engineering
Feature selection
PCA

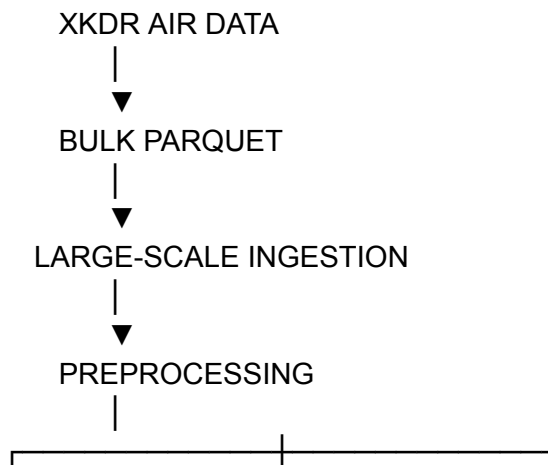
K-Means
DBSCAN
Anomaly detection
Regression
Classification
Forecasting
Model comparison
Explainability

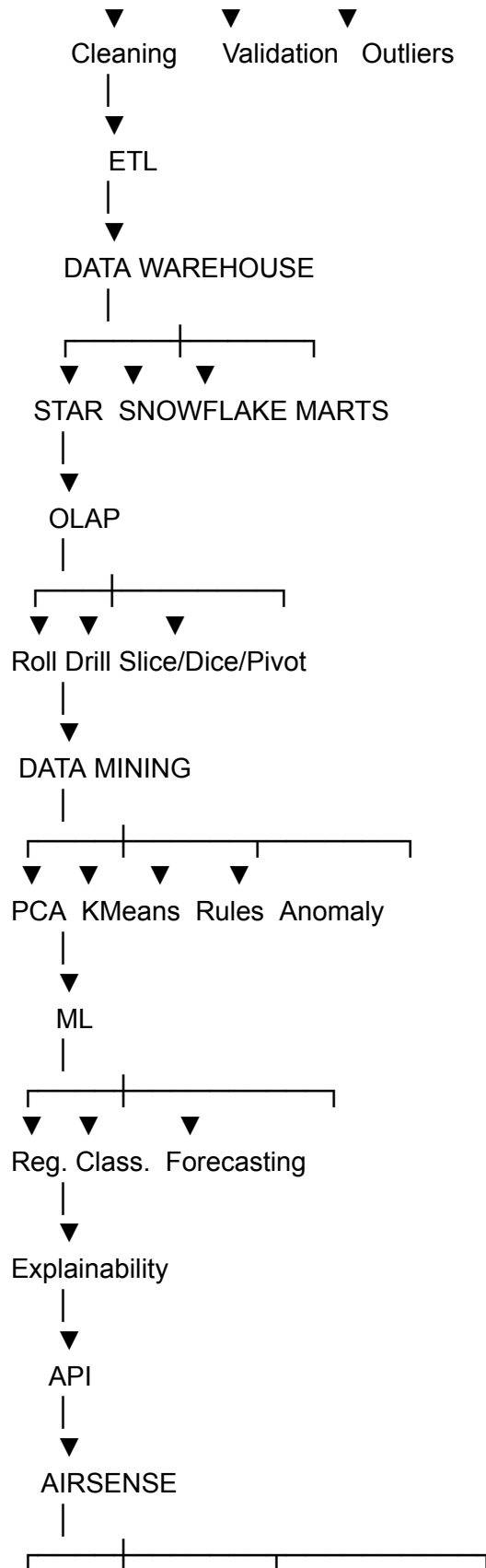
Member 3 — Full-Stack/UI

Next.js
Frontend
Maps
Charts
Dashboard
OLAP UI
ML UI
FastAPI
Database integration
Location permission
Deployment

All three contribute to one integrated system.

55. Final End-to-End Workflow





56. Minimum Required Feature Set

The project should not be considered complete without:

DWM

- ✓ Large-scale data source
- ✓ Data profiling
- ✓ Preprocessing
- ✓ ETL
- ✓ Fact tables
- ✓ Dimension tables
- ✓ Star schema
- ✓ Snowflake schema
- ✓ Time hierarchy
- ✓ Geographical hierarchy
- ✓ Data marts
- ✓ OLAP
- ✓ Roll-up
- ✓ Drill-down
- ✓ Slice
- ✓ Dice
- ✓ Pivot
- ✓ Aggregations
- ✓ Visualization

Data Mining / ML

- ✓ Feature engineering
- ✓ Feature selection
- ✓ Feature extraction
- ✓ PCA
- ✓ K-Means
- ✓ Anomaly detection
- ✓ Regression
- ✓ Classification
- ✓ Forecasting
- ✓ Model comparison

- ✓ Explainability

Product

- ✓ Citizen landing page
 - ✓ Plain-language AQI explanations
 - ✓ Location permission
 - ✓ Manual location fallback
 - ✓ Current/selected time
 - ✓ Activity selection
 - ✓ Pollution map
 - ✓ Historical analytics
 - ✓ Prediction
 - ✓ Anomaly analysis
 - ✓ DWM showcase
 - ✓ OLAP explorer
 - ✓ Feature lab
 - ✓ Model lab
 - ✓ Data-quality dashboard
 - ✓ ETL dashboard
-

57. Optional "Flashbang" Features

These are not required for the core project but would make the final demonstration significantly stronger.

Pollution Replay

Select:

Date: 15 Nov 2025

Time: 00:00

Press:

PLAY

and watch station pollution states change hour by hour across the map.

This visually demonstrates temporal warehouse data.

Pollution Journey

Select two locations and show:

Location A



Nearby stations



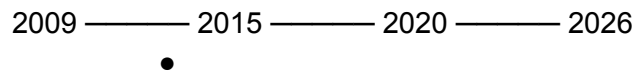
Historical pollution profile



Location B

AQI Timeline Scrubber

A timeline slider:



Move the slider and the entire dashboard updates.

OLAP Cube Visualizer

A 3D interactive representation:

Time

×

Location

×

Pollutant

with clicking/filtering to perform subcube exploration.

"Air Quality DNA"

Generate a visual pollution profile for a station:





and compare station profiles.

58. Academic Deliverables

The final submission should include:

- Project Report
- Research/Methodology
- Data Dictionary
- ETL Documentation
- Star Schema
- Snowflake Schema
- Warehouse SQL
- OLAP Queries
- Preprocessing Notebook
- ML Notebook
- Model Comparison
- Feature Selection
- Clustering Results
- Anomaly Results
- Visualization Results
- Web Application
- API Documentation
- Database Dump
- Requirements File
- README
- Presentation
- Demo Video

59. Source / Attribution Requirements

Cite:

XKDR Forum (2026), India Air Quality Database

and acknowledge:

Central Pollution Control Board CAAQM network

and

US Department of State monitors via AirNow

XKDR states that the compilation is released under **Creative Commons Attribution 4.0** and requires attribution to the database and its underlying sources.

Useful official/source resources:

[XKDR India Air Quality Database](#)

[XKDR Air Quality Database GitHub repository](#)

[XKDR Colab quickstart notebook repository](#)

[CPCB National Air Quality Index documentation](#)

60. Final Project Definition

Short title

AirSense

Full title

AirSense: A Large-Scale Urban Air Quality Data Warehouse and Machine Learning Based Intelligent Analytics Platform

One-line problem statement

To develop a large-scale citizen-facing air-quality intelligence platform that integrates historical air-quality observations into a multidimensional Data Warehouse, applies OLAP and data-mining techniques for spatiotemporal analysis, and uses Machine Learning for AQI forecasting, classification, clustering and anomaly detection to provide understandable environmental insights to users.

One-line USP

AirSense turns 196M+ historical air-quality observations into a single system where a citizen can understand current conditions while an evaluator can inspect the complete DWM + ML pipeline behind them.

The core philosophy

THE CITIZEN SEES:

"What's happening around me?"

THE DWM TEACHER SEES:

"196M observations → ETL → Warehouse → OLAP → Mining"

THE ML TEACHER SEES:

"Features → Selection/PCA → Regression → Classification
→ Clustering → Forecasting → Anomaly Detection
→ Model Comparison → Explainability"

YOUR TEAM SHOWS:

"All three are the same system."

That is the architecture I would lock **before starting the Colab work**. The most important design decision is to make the **warehouse and analytical marts the source of truth for the UI**, rather than exporting a pile of notebook screenshots and building an unrelated frontend. That way, the pages your team presents are direct visualizations of the actual work performed in Colab and PostgreSQL.